Customer Churn Prediction

Streamlit Dashboard Report

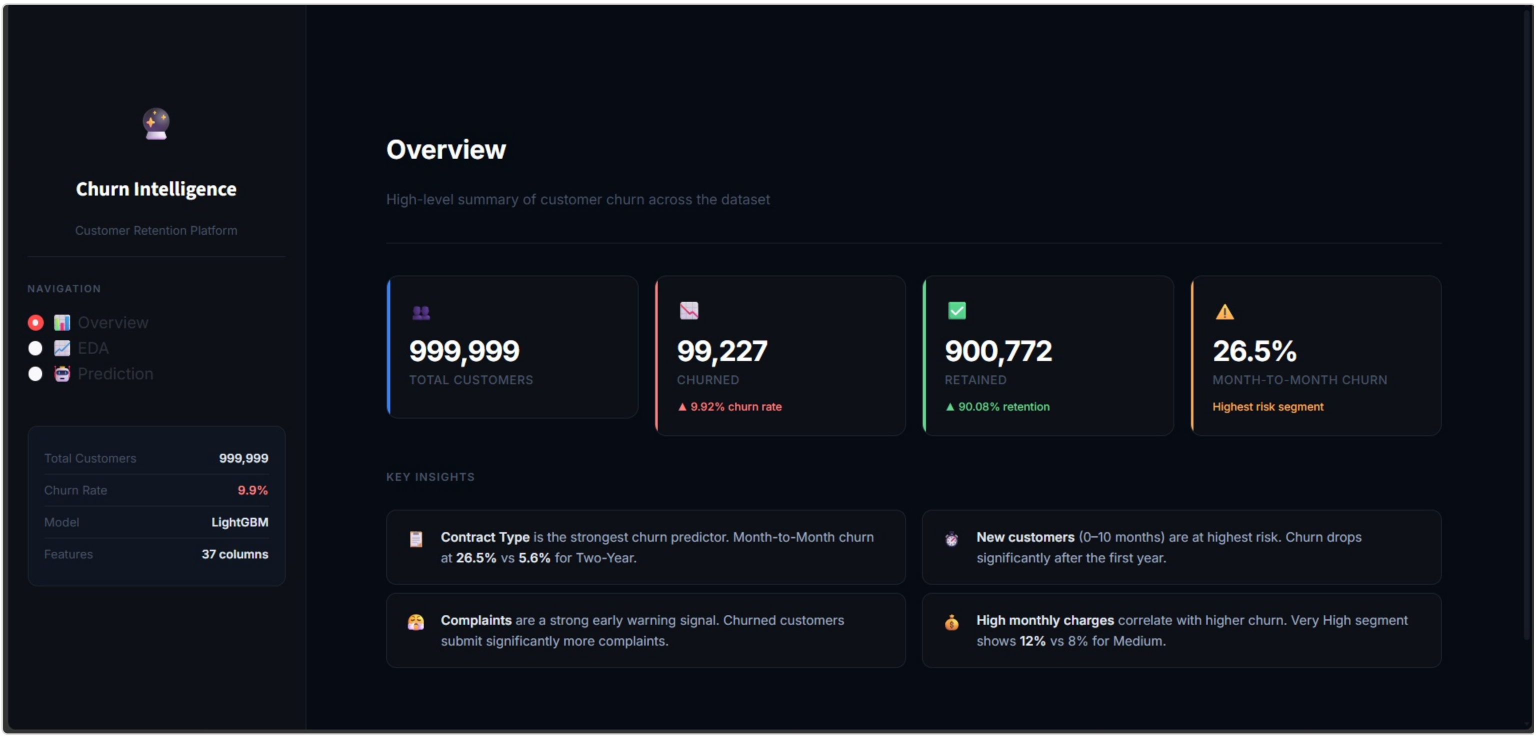
Deployment & Interactive Prediction Tool

Role: BI Developer / Deployment Engineer

Dataset: 999,999 Rows | Model: LightGBM | Features: 37

1. Dashboard Overview

The final phase of the project delivers an interactive Streamlit dashboard that brings together the data engineering, modeling, and analysis work into a single deployable application. The dashboard is organized into three pages: Overview, EDA, and Prediction, accessible through a fixed sidebar.



> 💡 The Overview page surfaces the four headline metrics — total customers, churned count, retained count, and the highest-risk segment (Month-to-Month contracts at 26.5%) — alongside the key insights extracted during EDA, so a non-technical viewer can grasp the churn story in seconds.

2. Dashboard Structure

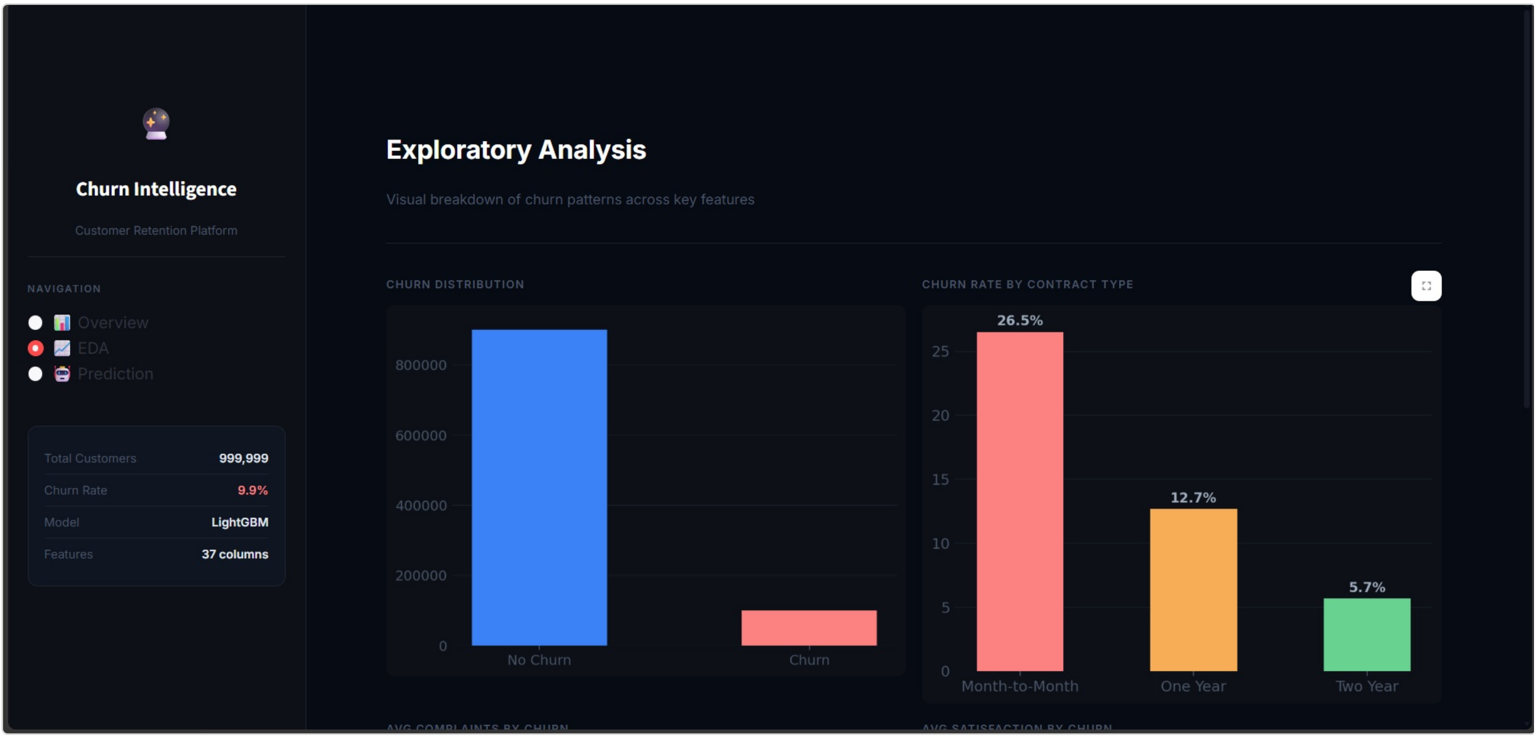
The app loads the cleaned dataset and the saved LightGBM model once (cached with `@st.cache_data` / `@st.cache_resource`) and reuses them across all three pages.

Page	Purpose	Key Elements
📊 Overview	High-level churn summary	Metric cards, key insights
📈 EDA	Visual breakdown of churn patterns	Distribution, contract type, complaints, satisfaction charts
🤖 Prediction	Live churn probability scoring	Interactive sliders, model inference, risk factors

> 💡 A dark theme (custom CSS injected via `st.markdown`) was used instead of Streamlit's default light theme to give the dashboard a more professional, analytics-product feel rather than a typical student notebook export.

3. EDA Page — Visual Breakdown

The EDA page reproduces the core findings from the exploratory analysis phase directly inside the dashboard, so stakeholders do not need to open a separate report to see the churn drivers.

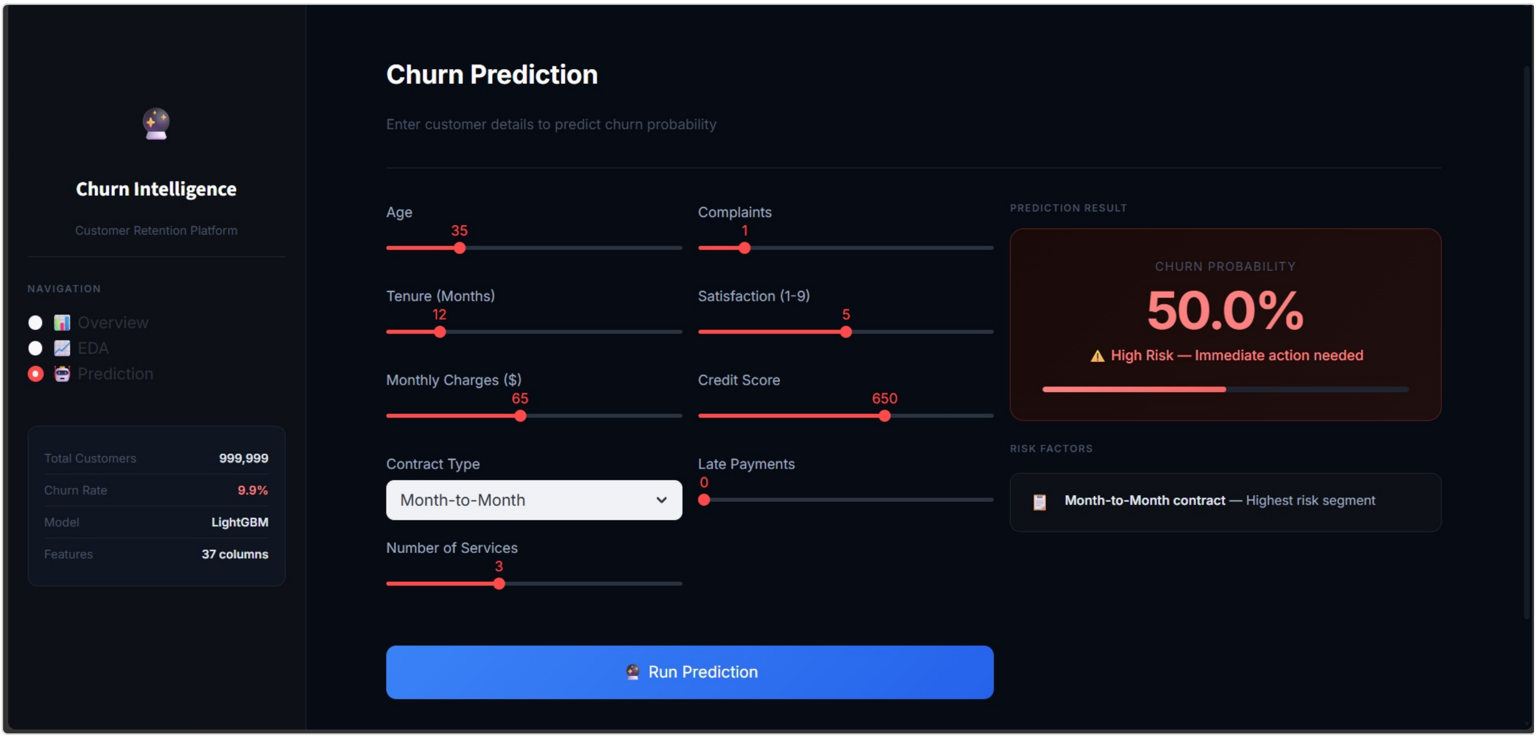


💡 Churn Distribution confirms the 9.9% imbalance. Churn Rate by Contract Type shows Month-to-Month customers churn at 26.5%, more than double the One-Year rate (12.7%) and over four times the Two-Year rate (5.7%).

💡 Average Complaints by Churn and Average Satisfaction by Churn both reinforce the same signal: churned customers file more complaints and report lower satisfaction on average than retained customers — consistent with the correlation analysis from the EDA phase.

4. Prediction Page — High-Risk Example

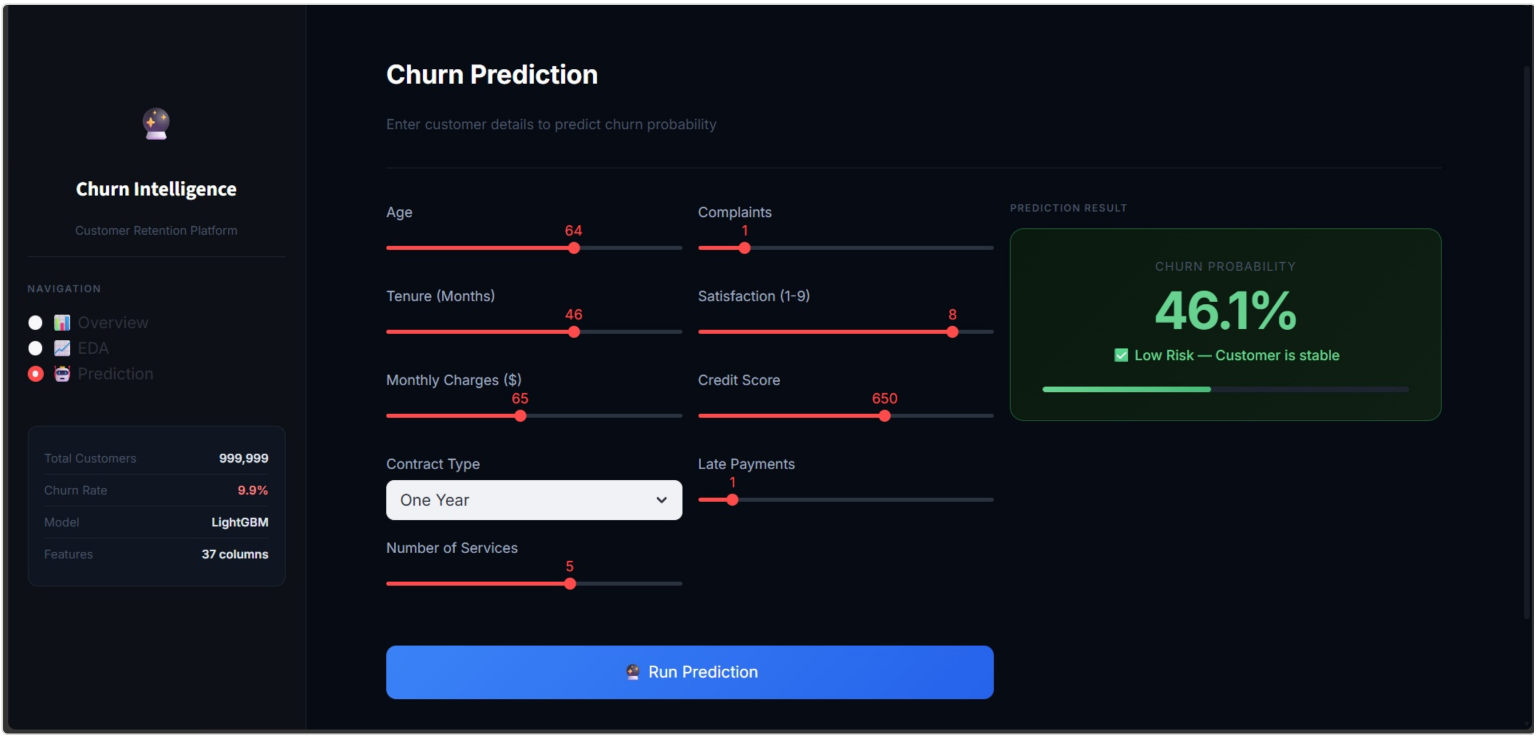
The Prediction page lets a user enter a customer's profile through sliders and a contract-type dropdown, then runs that profile through the trained LightGBM model to return a live churn probability with a risk label and contributing risk factors.



💡 Example: a 35-year-old customer on a Month-to-Month contract, 12 months tenure, $65 monthly charges, and average satisfaction returns a 50.0% churn probability, flagged as **High Risk** with "Month-to-Month contract — Highest risk segment" surfaced as the driving factor.

5. Prediction Page — Low-Risk Example

The same tool correctly differentiates a more stable customer profile, demonstrating that the model's output responds sensibly to changes in tenure, contract type, and satisfaction.



💡 Example: a 64-year-old customer on a One-Year contract, 46 months tenure, and a satisfaction score of 8/9 returns a 46.1% churn probability, labeled **Low Risk — Customer is stable**. The drop versus the high-risk example reflects the combined effect of longer tenure and a more committed contract type.

6. Technical Implementation Notes

A few practical issues came up while building and running the dashboard, documented here for reference when handing off the app.

- **File paths:** Streamlit's working directory depends on where `streamlit run` is launched from, not the script's location. Paths to the dataset and model were resolved relative to the project root using `os.path.dirname(os.path.abspath(__file__))` to avoid `FileNotFoundError` when running from different terminals.
- **Scaled features in the UI:** Since the saved model was trained on scaled data, the Prediction page sliders use raw, human-readable ranges (e.g. Age 18–90, Monthly Charges $20–120) and the input row is built directly into the model's expected 37-column format before calling `predict_proba`.
- **Caching:** `@st.cache_data` and `@st.cache_resource` were used so the 1M-row dataset and the model are loaded only once per session instead of on every interaction.
- **Styling:** the dark theme, metric cards, and chart color palette are implemented with injected CSS rather than third-party Streamlit themes, keeping the app self-contained in a single `streamlit_app.py` file.

7. Deliverables Summary

Deliverable	Description
streamlit_app.py	Full dashboard application (Overview, EDA, Prediction pages)
Overview Page	Headline churn metrics and key insight cards
EDA Page	Four interactive charts reproducing the core EDA findings
Prediction Tool	Live churn probability scoring with risk factor explanation

End of Streamlit Dashboard Report